

Smart Sensing Everywhere



RS6130

Ultra Low Power Single-Chip 60 GHz AiP Radar Sensor

with Wireless Connectivity (BLE5.3/802.15.4/Thread/Matter)

Datasheet

Document Sheets: 42

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1 Features

- **FMCW mmWave Radar**

- 58-64 GHz Frequency Range
- Up to 2.5cm Range Resolution
- 1 Transmit and 3 Receive Channels with AiP Antenna
- Integrated RFPLL, DPLL, Transmitter, Receiver, ADC and Baseband
- 17.0dBm Max TX EIRP Output Power
- 10dB Typical RX Noise Figure
- -95 dBc/Hz typical Phase Noise at 1MHz
- 10/20MHz IF Bandwidth
- Up to 400MHz/us Ramp Rate with <0.1% FM INL Error
- Configurable FMCW chirp profile
- Integrated Radar Baseband by supporting digital filter, frame control, FIFO and hardware accelerator(HWA)
- HWA for 1D/2D FFT, Static Clutter Remove and CFAR Operations

- **2.4GHz Wireless Connectivity**

- Protocols: Supports Bluetooth LE 5.3, Bluetooth Mesh, IEEE 802.15.4, Thread 1.3, Matter 1.1, and 2.4GHz proprietary
- Bluetooth LE PHY: 1Mbps, 2Mbps, Long Range S2 (500Kbps), S8 (125Kbps)
- TX Max Output Power: 10dBm@BLE mode
- RX Sensitivity: -95dBm @BLE 1Mbps

- **Application System**

- 32-bit RISC-V CPU with FPU (up to 128MHz) and I/D cache for Radar Post-processing, Wireless and Application
- Security Engine: Secure Boot, Efuse Key Protection, SRAM and XIP Decryption on the fly, TRNG, AES/SHA/RSA/ECC

- **Internal Memory**

- On Chip 384KB SRAM
- Configurable Radar Data Cube (up to 128KB)
- Configurable System RAM (up to 256KB)

- **Host and Device Interface**

- x3 UART
- x2 QSPI with XIP
- x2 I2C
- x8 PWM
- x10 GPADC

- **Clock Source**

- 32/40MHz Crystal Oscillator for Primary Clock Reference
- Internal Low Power LSI RC Oscillator
- Internal High Speed MSI RC Oscillator
- Optional 32kHz Crystal for Wireless Low Power Operation

- **Power Management**

- Single Power Supply from 1.7V to 5.5V
- Built-in LDO Network for Enhanced PSRR
- BOM-Optimized and Power-Optimized Modes

- **Ultra Low Power Property**

- Power States: Shutdown, Standby, Idle and Active (Sensing, Processing and Connecting)
- Fast Wakeup from Standby to Active
- Motion&Presence Detection: <0.1mW
- BLE Advertising&Connecting: <0.1mW

- **Package**

- AiP FCCSP: 8.8 x 5.8 mm² with 56pins

- **Temperature Range**

- Operating Junction Temperature Range: -40°C to 105°C

2 Applications

- Motion detection
- Presence detection
- Gesture recognition
- Video doorbell
- IP camera
- Air conditioner
- Refrigerators
- Lighting
- PC/Notebooks
- Televisions
- Tablets
- Wearable

3 General Description

3.1 Overview

The RS6130 is a single chip 60GHz 1T3R FMCW AiP radar sensor with multi-protocol wireless connectivity (BLE5.3/802.15.4/Thread/Matter) inside. The device is portioned into four subsystems, shown in Figure 3-1:

- **mmWave Radar Subsystem:** This block includes all high-performance radar RF/analog and baseband. It also includes the chirp generator and tx-to-rx finite state machine. HWA is included in baseband, used to offloading specified radar processing such as 1D/2D FFT and CFAR.
- **2.4GHz Wireless Connectivity Subsystem:** This block contains whole 2.4G multi-protocol wireless RF, Mod/Demod, Baseband Protocol, and the data interacts with application CPU by internal data bus.
- **Application Subsystem:** It mainly includes one 32-bit RISC-V CPU with FPU and I/D cache. CPU connects SRAM, peripheral interface, and other subsystems by AHB and APB bus. RISC-V CPU is used to run radar post-processing algorithm, wireless protocol and other system tasks.
- **Power and Clock Subsystem:** This chip is complex single chip SoC with multiple function systems, we use single power and clock subsystem to do centralized management. It includes several clock sources (including LSI, MSI, DCXO and SPLL), power blocks(LDOs, BG and Clamp) and temperature sensor.

RS6130 is specifically designed to feature the capability of ultra low power radar sensing. It contains four main power states for low power management: Shutdown, Standby, Idle and Active. Our EFSENS™ ultra low power technology enable the sensor to keep active time more than years feed by button battery. Its configuration and data acquisition are enabled with UART/SPI/I2C digital interface with other devices.

3.2 Block Diagram

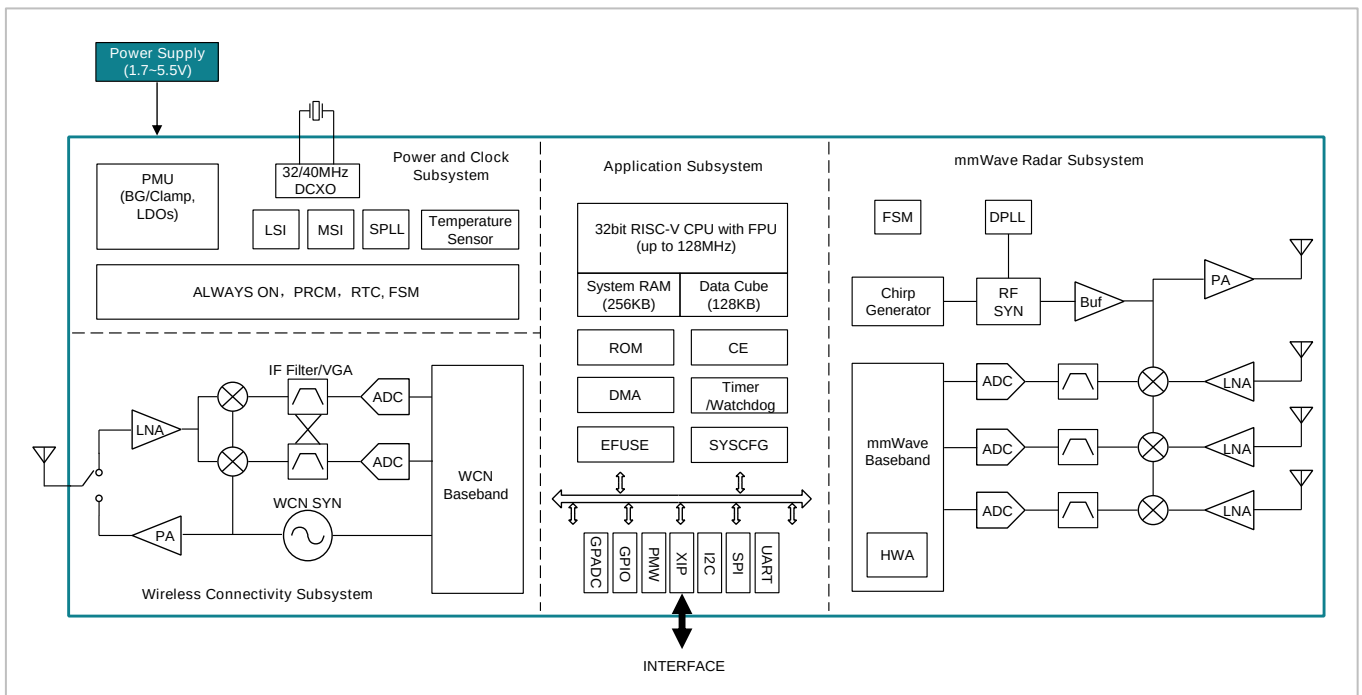


Figure 3-1 Block Diagram

4 Pin Configurations

4.1 Pin Diagrams

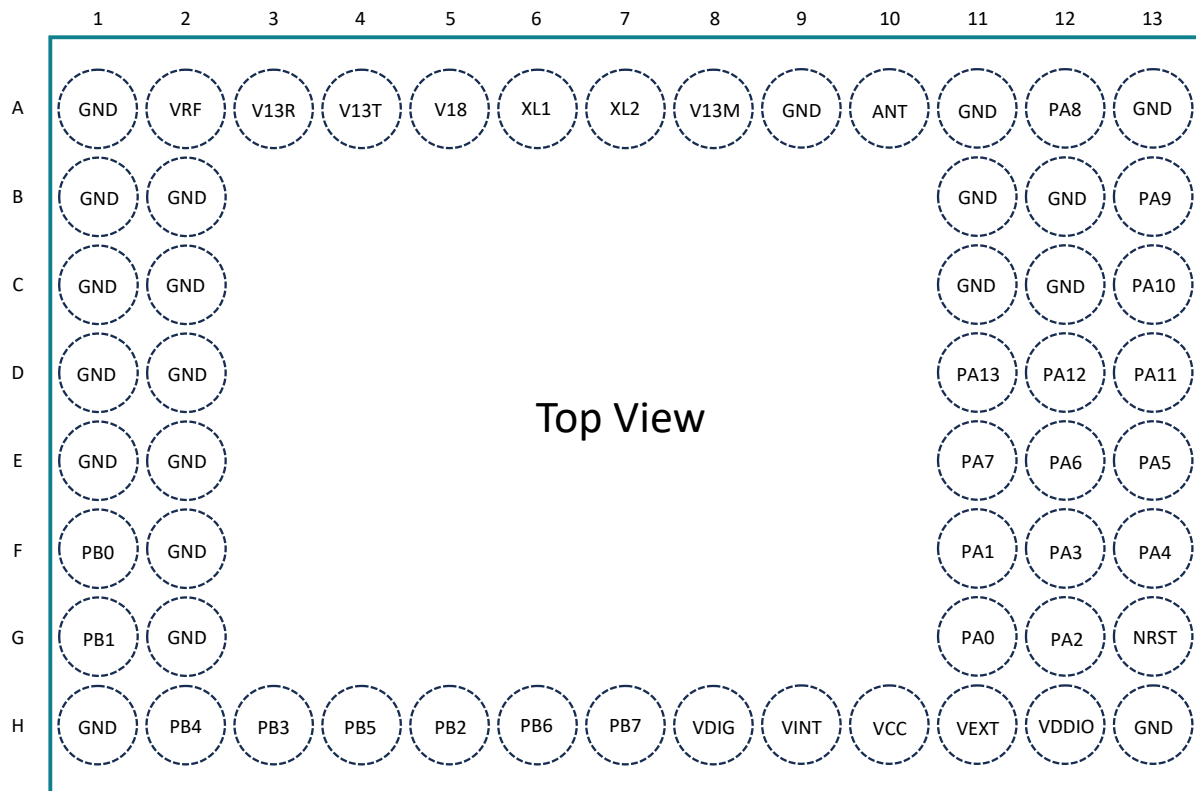


Figure 4-1 Pin Diagrams

4.2 Pin Descriptions

Table 4-1 Pin Descriptions

Pin Name	I/O	Pin No.	Pin Description
VCC	Power In	H10	PMU power supply, 1.7~5.5V
VINT	Power Out	H9	PMU INT LDO output, Default 1.8V, 0.9~2.5V/0.1V step
VEXT	Power Out	H11	PMU EXT LDO output, Default 3.3V, 1.6~3.3V/0.1V step
VDIG	Power Out	H8	PMU DIG LDO output, Default 1.15V, 0.65~1.35V/0.05V step
VDDIO	Power In	H12	GPIO Power supply, 1.7~3.6V
V18	Power In	A5	MMW Analog power supply, 1.5~2.5V
V13T	Power In	A4	MMW TX power supply, 1.2~2.5V
V13R	Power In	A3	MMW RX power supply, 1.2~2.5V
VRF	Power Out	A2	MMW RF LDO output, default 1.0V

V13M	Power In	A8	2.4G wireless analog power supply, 1.2~2.5V
XL1	Analog	A6	Crystal oscillator output
XL2	Analog	A7	Crystal oscillator input
ANT	RF	A10	2.4G RF input and output
GND	Ground	A1,B1,C1,D1,E1, H1,B2,C2,D2,E2, F2,G2,A9,A11,A 13,B11,B12,C11, C12,H13	Chip ground
NRST	In	G13	Chip reset, need feed VDDIO to power supply
PA0	In/Out	G11	Programmable GPIO input/output, Wakeup IO
PA1	In/Out	F11	Programmable GPIO input/output, Wakeup IO
PA2	In/Out	G12	Programmable GPIO input/output, Wakeup IO
PA3	In/Out	F12	Programmable GPIO input/output, Wakeup IO
PA4	In/Out	F13	Programmable GPIO input/output, Wakeup IO
PA5	In/Out	E13	Programmable GPIO input/output, Wakeup IO
PA6	In/Out	E12	Programmable GPIO input/output, Wakeup IO
PA7	In/Out	E11	Programmable GPIO input/output, Wakeup IO
PA8	In/Out	A12	Programmable GPIO input/output, Wakeup IO
PA9	In/Out	B13	Programmable GPIO input/output, Wakeup IO
PA10	In/Out	C13	Programmable GPIO input/output, Wakeup IO
PA11	In/Out	D13	Programmable GPIO input/output, Wakeup IO
PA12	In/Out	D12	Programmable GPIO input/output, Wakeup IO
PA13	In/Out	D11	Programmable GPIO input/output, Wakeup IO
PB0	In/Out	F1	Programmable GPIO input/output, Wakeup IO
PB1	In/Out	G1	Programmable GPIO input/output, Wakeup IO
PB2	In/Out	H5	Programmable GPIO input/output
PB3	In/Out	H3	Programmable GPIO input/output
PB4	In/Out	H2	Programmable GPIO input/output
PB5	In/Out	H4	Programmable GPIO input/output
PB6	In/Out	H6	Programmable GPIO input/output
PB7	In/Out	H7	Programmable GPIO input/output

4.3 GPIO Multiplexing

Table 4-2 GPIO Multiplexing

GPIO Name	Func2	Func3	Func4	Func5	Func6	Func7	Func after rst
PA0/WIO0	SPIO_CS	UART1_TX			PWM0		OFF/No-Pull
PA1/WIO1	SPIO_CLK	UART1_RX			PWM1		OFF/No-Pull
PA2/WIO2	SPIO_MOSI	UART1_RTS	I2C0_SDA	UART2_TX	PWM2		OFF/No-Pull
PA3/WIO3	SPIO_MISO	UART1_CTS	I2C0_SCL	UART2_RX	PWM3		OFF/No-Pull
PA4/WIO4	SPIO_HOLD	I2C0_SDA			PWM4		OFF/No-Pull
PA5/WIO5	SPIO_WP	I2C0_SCL			PWM5		OFF/No-Pull
PA6/WIO6	UART1_RTS	UART2_TX	I2C1_SDA		PWM6	GPADC6	OFF/No-Pull
PA7/WIO7	UART1_CTS	UART2_RX	I2C1_SCL		PWM7	GPADC7	OFF/No-Pull
PA8/WIO8/LXL1 ¹	UART1_TX	I2C1_SCL			PWM0	GPADC0	OFF/No-Pull
PA9/WIO9/LXL2 ¹	UART1_RX	I2C1_SDA			PWM1	GPADC1	OFF/No-Pull
PA10/WIO10	I2C1_SDA				PWM2	GPADC2	OFF/No-Pull
PA11/WIO11	I2C1_SCL				PWM3	GPADC3	OFF/No-Pull
PA12/WIO12	UART2_RX	UART1_RTS			PWM4	GPADC4	OFF/No-Pull
PA13/WIO13/TEST ²	UART2_TX	MCO_CLK			PWM5	GPADC5	IN/UP
PB0/WIO16	UART0_TX				PWM0		OFF/UP
PB1/WIO17	UART0_RX				PWM1		OFF/UP
PB2	UART0_RTS	FC_WP	SPIO_WP		PWM2		OFF/UP
PB3	UART0_CTS	FC_HOLD	SPIO_HOLD		PWM3		OFF/UP
PB4	FC_CS	UART2_RX	SPIO_CS		PWM4		OFF/No-Pull
PB5	FC_CLK	UART2_TX	SPIO_CLK		PWM5		OFF/No-Pull
PB6	FC_MOSI	I2C0_SDA	SPIO_MOSI		PWM6		OFF/No-Pull
PB7	FC_MISO	I2C0_SCL	SPIO_MISO		PWM7		OFF/No-Pull

1. LXL1 and LXL2 are 32K external crystal pins, they share the same ball with PA8 and PA9, when used as analog pin, GPIO function should be disabled.
2. When chip is first power on or after reset, PA13 will be configured to input and pull-up mode, if it is tie to GND then will go into ATE test mode.
3. The Chip use uart0 as default interface for boot and firmware upgrade.

5 AiP Antenna

5.1 Antenna Configuration

Figure 5-1 shows the antenna configuration of the RS6130. It has one transmitter antenna and three receiver antennas. The RX antenna spacing is 2.45mm for both azimuth and elevation.

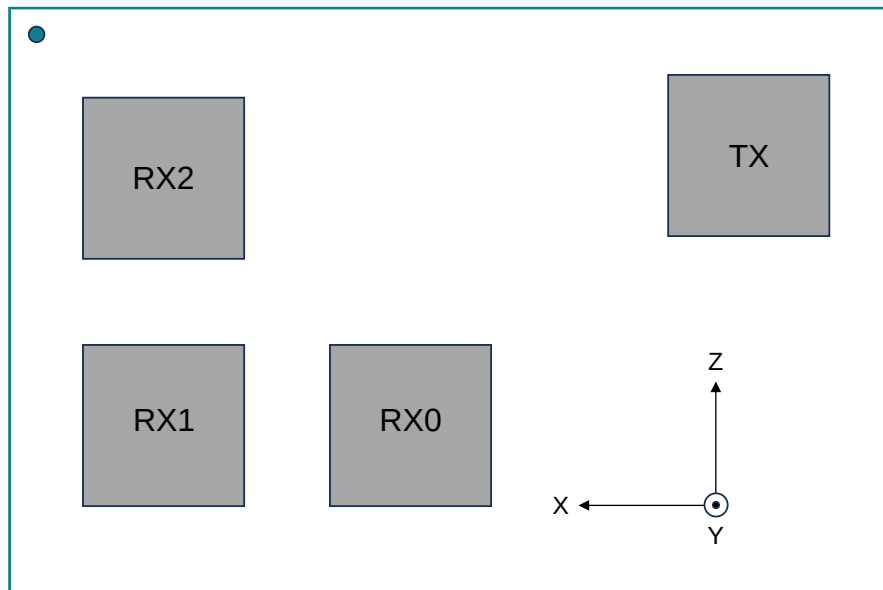


Figure 5-1 Antenna Configuration (Top View)

5.2 Antenna Radiation Patterns

Figure 5-2 shows the antenna radiation pattern with regard to azimuth and Figure 5-3 shows the antenna radiation pattern with regard to elevation for TX. Both show the radiation pattern for TX and RX0, RX1 and RX2 together. All of the measurements were done with a TX and RX combination together. Thus, for the -6dB beam width, you must see a -12db (Tx (-6dB) + Rx(-6dB)) number.

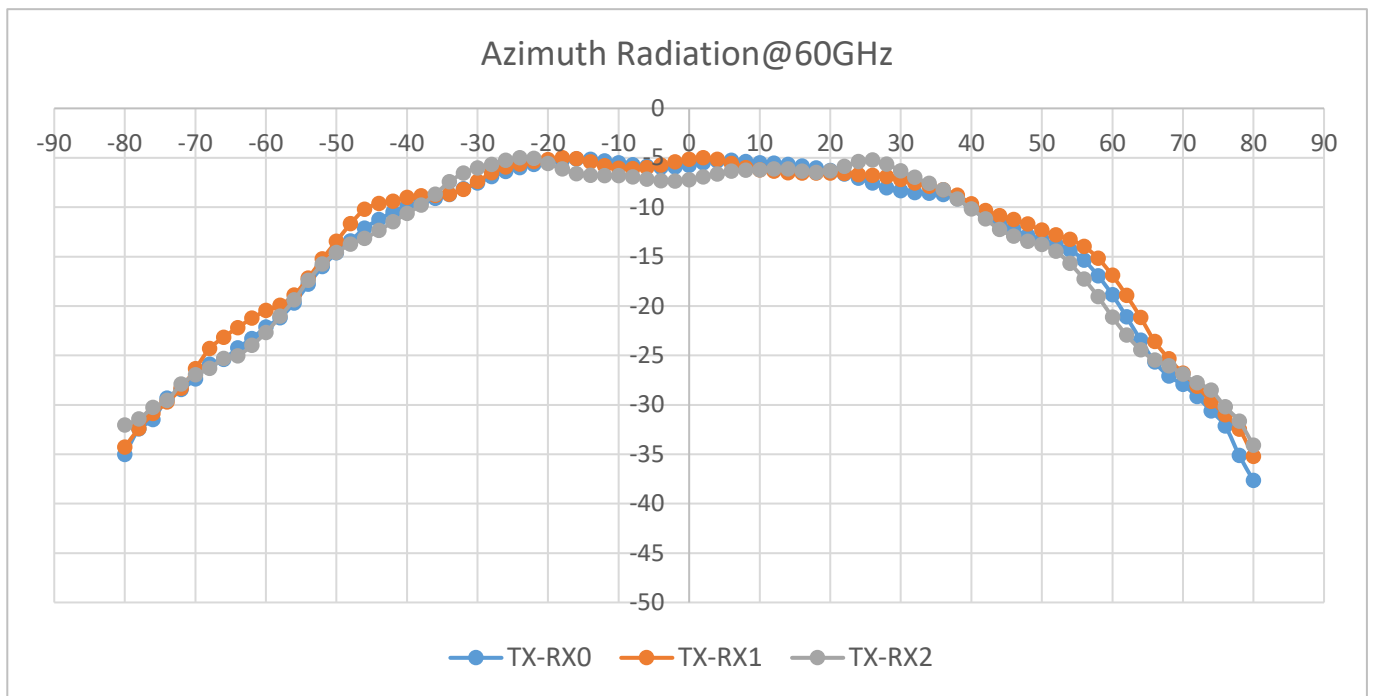


Figure 5-2 Antenna Azimuth(XOZ) Radiation Pattern^{1,2}

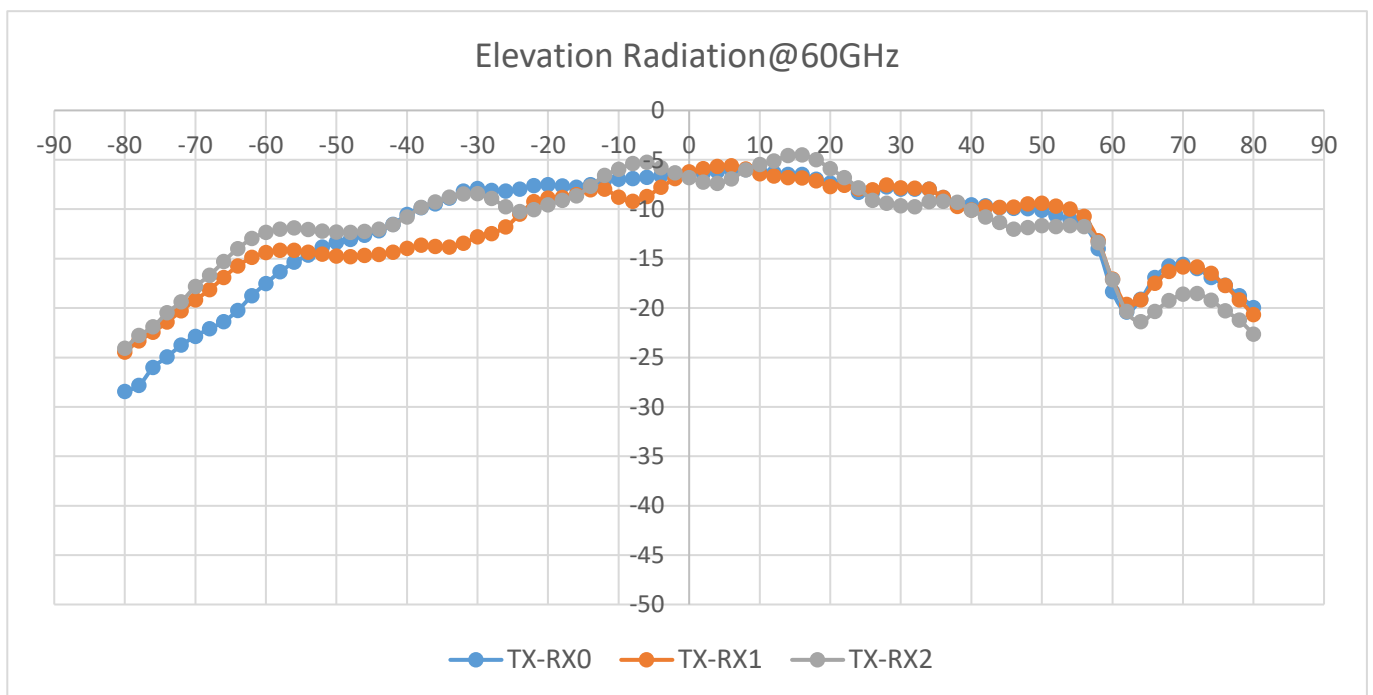


Figure 5-3 Antenna Elevation(YOZ) Radiation Pattern^{1,2}

1. Radiation Patterns are related with PCB drawing, the data is based our internal evaluation PCB board.
2. Horizontal axis: degree; Longitudinal axis: dB;

6 Electrical Specifications

6.1 Absolute Maximum Ratings

Table 6-1 Absolute Maximum Ratings

Parameters ¹		Min.	Max.	Unit
VCC	PMU power supply	-0.30	6.50	V
VINT	PMU INT LDO output	-0.30	3.80	V
VEXT	PMU EXT LDO output	-0.30	3.80	V
VDIG	PMU DIG LDO output	-0.30	1.40	V
VDDIO	GPIO Power supply	-0.30	3.80	V
V18	MMW Analog power supply	-0.30	3.80	V
V13T	MMW TX power supply	-0.30	3.80	V
V13R	MMW RX power supply	-0.30	3.80	V
VRF	MMW RF LDO output	-0.30	1.40	V
V13M	2.4G wireless analog power supply	-0.30	3.80	V
V _{XL}	Crystal oscillator input and output voltage	-0.30	2.70	V
P _{ANT}	2.4G RF input power	-	10	dBm
V _{IO}	GPIO voltage for input and output	-0.30	3.80	V
I _{IO}	GPIO In/Out current for input and output	-100	100	mA
T _{opr}	Ambient Operating Temperature	-40	105	°C
T _{junction}	Operating junction temperature range	-40	125	°C
T _{stg}	Storage temperature range after soldered onto PC board	-55	150	°C
V _{HBM}	Human-body model (HBM), ANSI/ESDA/JEDEC JS-001	-2000	2000	V
V _{CDM}	Charged-device model (CDM), ANSI/ESDA/JEDEC JS-002 ²	-500	500	V

1. Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under Recommended Operating Conditions is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
2. Corner pins are rated as ± 750 V.

6.2 Recommended Operating Conditions

Table 6-2 Recommended Operating Conditions

Parameters	Min.	Max.	Unit
------------	------	------	------

VCC	PMU power supply	1.70	5.50	V
VINT	PMU INT LDO output	1.20	2.50	V
VEXT	PMU EXT LDO output	1.60	3.60	V
VDIG	PMU DIG LDO output	0.65	1.25	V
VDDIO	GPIO Power supply	1.70	3.60	V
V18	MMW Analog power supply	1.60	2.50	V
V13T	MMW TX power supply	1.20	2.50	V
V13R	MMW RX power supply	1.20	2.50	V
V13M	2.4G wireless analog power supply	1.20	2.50	V
VRF	MMW RF LDO output	0.80	1.20	V

6.3 Digital IO Characteristics

Table 6-3 DC Characteristics

Parameters		Min.	Max.	Unit
VIL	Input Low Voltage	-0.3	0.35*VDDIO	V
VIH	Input High Voltage	0.65*VDDIO	1.1*VDDIO	V
VOL	Output Low Voltage	-0.3	0.1*VDDIO	V
VOH	Output High Voltage	0.9*VDDIO	1.1*VDDIO	V
RPU	Input Pull-up Resistance@VDDIO=3.3V	35	75	Kohm
	Input Pull-up Resistance@VDDIO=1.8V	70	150	Kohm
RPD	Input Pull-down Resistance@VDDIO=3.3V	30	75	Kohm
	Input Pull-down Resistance@VDDIO=1.8V	60	150	Kohm

6.4 mmWave Radar RF Specifications

Table 6-4 mmWave Radar RF Specifications

Parameters		Min.	Typ.	Max.	Unit
Receiver	Noise figure	-	10.0	-	dB
	Maximum gain	-	51	-	dB
	Gain range	-	24	-	dB
	Gain step size	-	6	-	dB
	IF bandwidth ¹	10	-	20	MHz
	ADC resolution	-	12	-	Bit

	Gain mismatch variation (over temperature)		-	+/-0.5	-	dB
	Phase mismatch variation (over temperature)		-	+/-3	-	°
Transmitter	Max EIRP output Power		-	17.0	-	dBm
	Power backoff range		-	20	-	dB
	Temperature sensor accuracy		-	+/-3	-	°C
Synthesizer	Frequency range		58	-	64	GHz
	Continuous linear sweep frequency bandwidth		-	-	6	GHz
	Ramp rate		-	-	400	MHz/us
	Phase noise at 1MHz offset ²	2GHz sweep BW	-	-95	-	dBc/Hz
		6GH sweep BW	-	-92	-	dBc/Hz

1. IF bandwidth can be configured to 10M/20M mode, and LPF is also be configured with IF bandwidth.
2. Data is measured at 60GHz with DCXO at 40MHz at typical 25°C. MMW Synthesizer can be configured to high BW mode (6GHz sweep bandwidth) or high performance mode (better phase noise, 2GHz sweep bandwidth).

6.5 Wireless Connectivity RF Specifications

Table 6-5 Wireless Connectivity RF Specifications

Parameters		Min.	Typ.	Max.	Unit
Receiver	Frequency range	2400	-	2484	MHz
	RX sensitivity@BLE-1Mbps	-	-95	-	dBm
	RX sensitivity@BLE-2Mbps	-	-92	-	dBm
	RX sensitivity@BLE-1Mbps, Coded2	-	-99	-	dBm
	RX sensitivity@BLE-1Mbps, Coded8	-	-101	-	dBm
	RX sensitivity@IEEE802.15.4	-	-100	-	dBm
	Maximum receiving power	-	0	-	dBm
	BLE C/I@Co-channel	-	7	-	dB
	BLE C/I@+1MHz ¹	-	-2	-	dB
	BLE C/I@+2MHz	-	-37	-	dB
	BLE C/I@+3MHz	-	-41	-	dB
	BLE C/I@+4MHz	-	-41	-	dB
	BLE C/I@-1MHz	-	2.5	-	dB
	BLE C/I@-2MHz	-	-19	-	dB
	BLE C/I@-3MHz	-	-25	-	dB
	BLE C/I@-4MHz	-	-37	-	dB
	IEEE802.15.4 C/I@+10MHz ²	-	-59	-	dB

	IEEE802.15.4 C/I@+5MHz		-	-54	-	dB
	IEEE802.15.4 C/I@-5MHz		-	-40	-	dB
	IEEE802.15.4 C/I@-10MHz			-60	-	dB
Transmitter	Frequency range		2400	-	2484	MHz
	FSK data rate	BLE	-	1.0	2.0	Mbps
		802.15.4	-	2.0	-	Mbps
	Modulation deviation	BLE 1Mbps	225	250	275	kHz
		BLE 2Mbps	450	500	550	kHz
		IEEE 802.15.4	-	500	-	kHz
	Output power range		-20	-	11.5	dBm
	In band spur. emission at 1Mbps@2M offset ³		-	-50	-	dBm
	In band spur. emission at 1Mbps@>=3M offset ³		-	-54	-	dBm
	In band spurious emission at 2Mbps@4M offset ³		-	-55	-	dBm
	In band spur. emission at 2Mbps@5M offset ³		-	-57	-	dBm
	In band spur. emission at 2Mbps@>=6M offset ³		-	-60	-	dBm
	Out band spur. emission at 4dBm output power	<2.4GHz	-	-56	-	dBm
		2.48~12GHz (wo harmonics)	-	-56	-	dBm
	Harmonics at 4dBm output power	2 nd	-	-41	-	dBm
		3 th	-	-43	-	dBm

1. The desired signal is at -67dBm;
2. The desired signal is at -82dBm;
3. Output power at 2dBm;

6.6 Power Specifications

6.6.1 Power Topologies

The chip supports two unique power topologies for BOM Optimized and Power Optimized modes. BOM Optimized Mode uses internal LDOs of PMU only to reduce extra device cost. Power Optimized Mode uses external DCDC to feed to V13 to reduce power consumption. The main power supply VCC can be provide from 1.7 to 5.5V depends on the system request.

VDDIO can be provide by EXT LDO inside chip or from outside. When Power Optimized Mode is selected, the EXT LDO is provided for V18 by default and VDDIO will be fed by external power supply. When BOM Optimized Mode is selected, the EXT LDO is provided to VDDIO by default, and also can be used for V18 power supply.

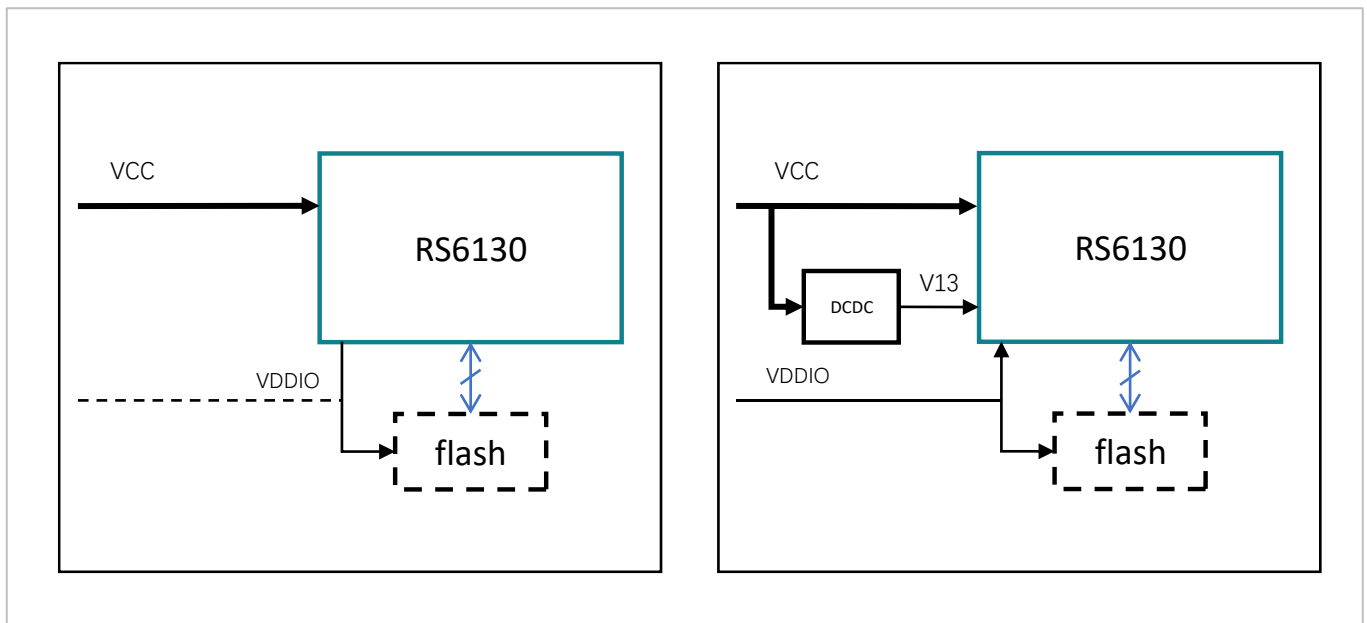


Figure 6-1 BOM Optimized (Left) and Power Optimized (Right) Mode

6.6.2 Power Save Modes

Table 6-6 Power States

Parameters		
Active	Sensing	Radar power state when RF/chirp is ongoing
	Processing	Radar data processing state when RF on is finished
	Connecting	Wireless active power state when RF/PHY /MAC is on
Idle		Power state during inter-frame/inter-chirp idle time
Standby		Lowest power state of the chip(only RTC power domain is activated) where the contents can be retained by retention SRAM and chip can fast wakeup to active
Shutdown		Chip is power down by NRST pin setting to low

6.6.3 Typical Power Consumption

Table 6-7 Typical Power Consumption at VCC=3.3V¹

Parameters		Power Optimized	BOM Optimized	Unit
Radar Sensing Mode (Motion&Presence Detection at 1Hz)	RF front end configuration : 1TX, 1RX Number of frame per second = 1 1D people detecting distance: 6~8m	18	29	uA

Radar Sensing Mode (Motion&Presence Detection at 2Hz)	RF front end configuration : 1TX, 3RX Number of frame per second = 2 1D/2D/3D people detecting distance: 6~8m	35	51	uA
Radar Sensing Mode (Motion&Tracking Detection at 4Hz)	RF front end configuration : 1TX, 3RX Number of frame per second = 4 1D/2D/3D people detecting distance: 12~16m	230	300	uA
BLE Connecting Mode	BLE connecting mode with 1s period at 0dBm output power Radar is Standby	18	-	uA
BLE Advertising Mode	BLE advertising mode with 1s period at 0dBm output power Radar is Standby	25	-	uA

1. The power is estimated based on initial silicon measurements and might be subjected to change/improvement pertaining to further characterization.

7 Function Description

7.1 Overview

The RS6130 is a fully integrated SoC which includes mmWave radar subsystem, programmable CPU application subsystem, 2.4GHz multi-protocol wireless connectivity subsystem and power/clock subsystem, as shown in Figure 7-1 System Block Diagram. It is designed for Artificial Intelligence and Internet of Things (AIOT) sensing applications, such as motion sensing, gesturing, automation sensing and building energy saving.

The SoC is optimized for low-power operation by using several low-power states and fast wake-up times from hardware to software. Multiple power domains and clocks can be shut down individually. The application subsystem, mmWave subsystem and wireless subsystem can be put into low-power states independently, supporting a variety of application use cases.

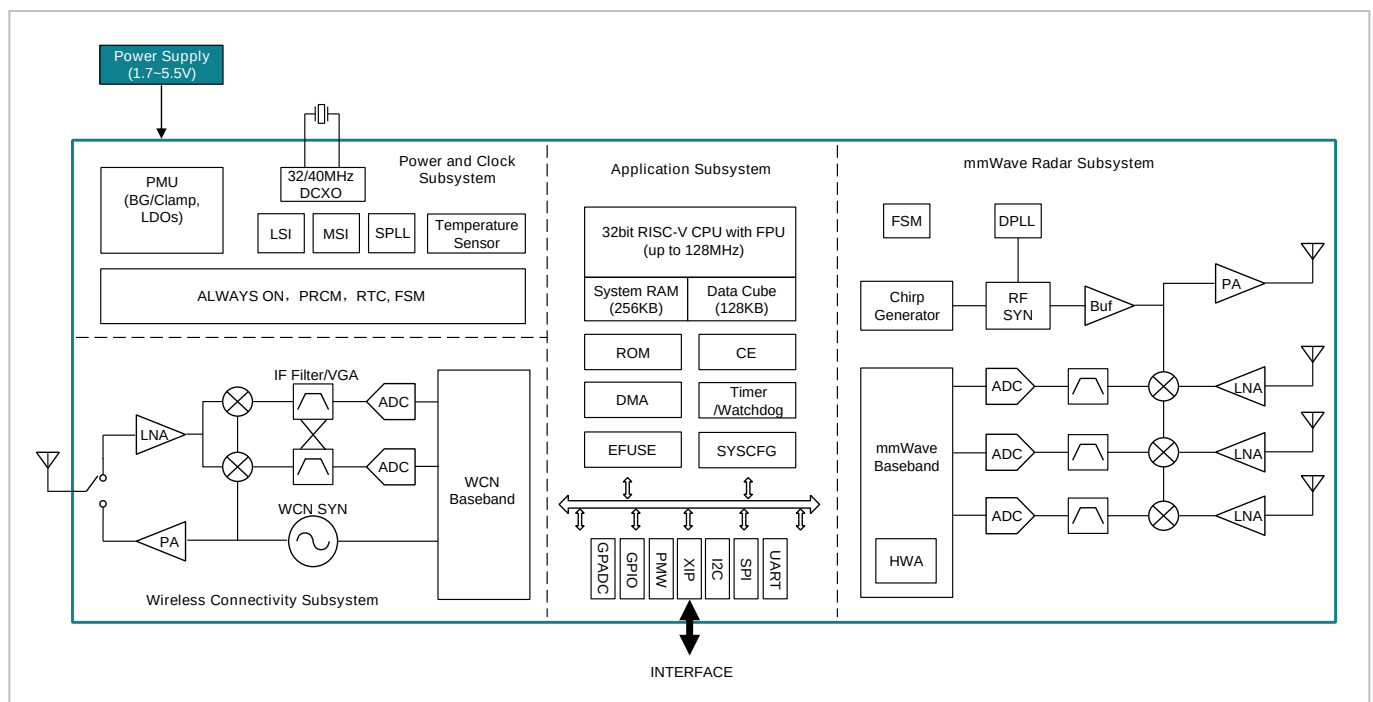


Figure 7-1 System Block Diagram

7.2 mmWave Radar Subsystem

The mmWave radar subsystem includes receiver, transmitter, synthesizer, chirp generator and baseband. It is designed to transmit and receive FMCW radar signal, and further to transfer radar signal to desired target information, such as range, velocity, angle and micro-motion.

7.2.1 Receiver

Receiver block consists of three parallel channels. A single receive channel consists of an LNA, mixer, IF filtering, ADC conversion, and decimation. All three receive channels can either operate simultaneously or can be powered down individually based on system power needs and application design. The RS6130 band-pass IF chain has configurable lower cutoff frequencies above 500 kHz and can support bandwidths up to 20 MHz.

7.2.2 Transmitter

Transmitter block consists of single transmit chain with amplitude control for power back-off. A power stage driving by driver stage to get high power(>10dBm) for long range detection.

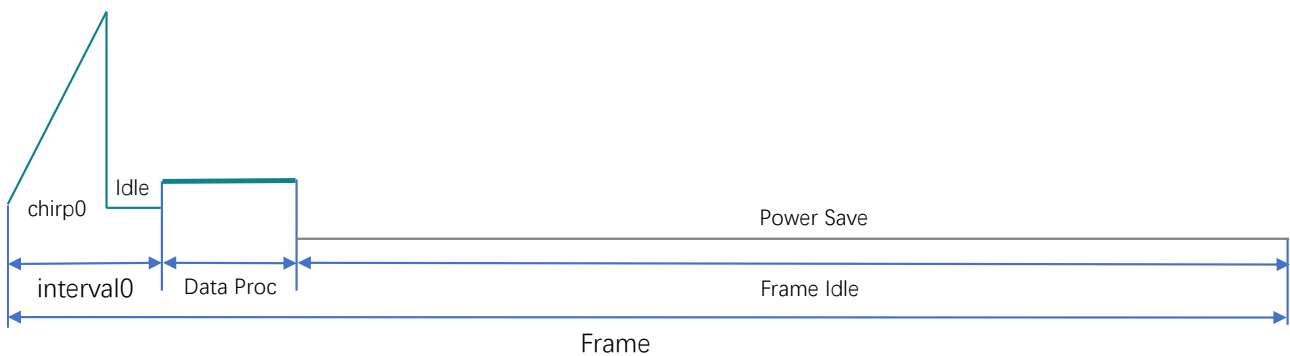
7.2.3 Synthesizer

Synthesizer block generates 58 to 64 GHz from an input reference from a crystal. It has a built-in oscillator circuit followed by a DPLL and a RF synthesizer circuit. The output of the RF synthesizer is then processed by an X4 multiplier to create the required frequency in the 57 to 66 GHz spectrum. The RF synthesizer output is modulated by the timing engine block to create the required waveform for effective sensor operation.

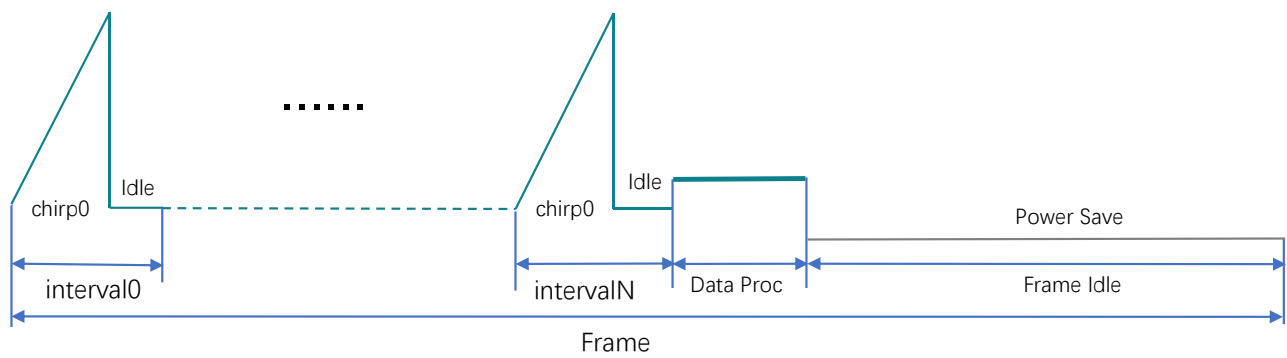
7.2.4 Chirp Generator

Chirp generator block is used to generate different kinds of FMCW chirps by supporting for chirps, intervals and frames. The device only support ramp up FMCW mode to simplified system. The chirp profile is shown in Figure 7-2 , We can set three different frame mode:

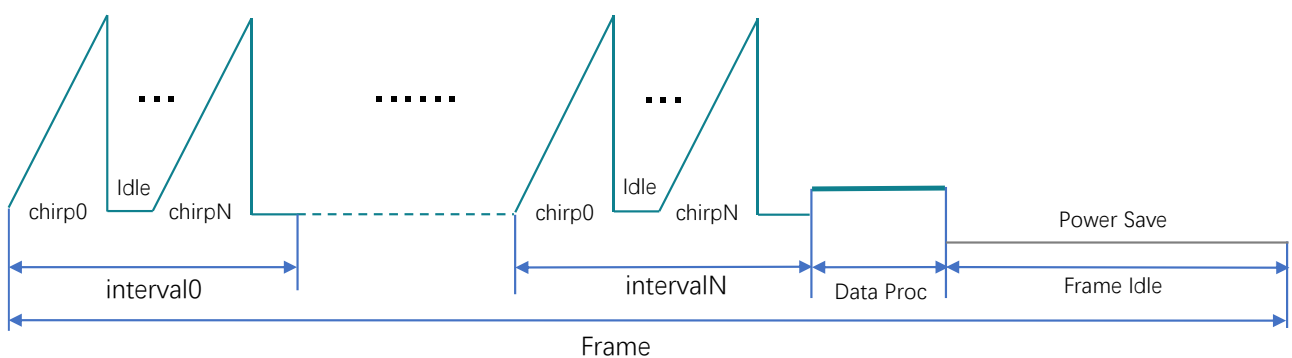
- (a) **Single-Interval and Single-Chirp Frame Mode:** Only one chirp in a frame, 1D-FFT with range information is obtained only, and system can go into Standby at frame idle time to significantly save power;
- (b) **Multi-Interval with Single Chirp in Interval Frame Mode:** One frame with multiple interval for 2D-FFT calculation to get both range and velocity information, interval number represents the Doppler number;
- (c) **Multi-Interval with Multi-Chirp Accumulation Frame Mode:** Based on the above mode (b), the chirp number can be set to $N > 1$ at each interval, which will set chirp to accumulation (averaging) mode every interval. Its possible to across closely spaced chirps to reduce memory requirement.



(a) Single-Interval and Single-Chirp Frame Mode



(b) Multi-Interval with Single Chirp in Interval Frame Mode



(c) Multi-Interval with Multi-Chirp Accumulation Frame Mode

Figure 7-2 Chirp Profile of Frame

7.2.5 Baseband

Baseband block consists of several signal processing units, such as digital filter, frame control, FIFO and hardware accelerator(HWA). HWA is used to offloading common radar processing such as FFT, Constant False Alarm rate (CFAR) and pre-processing. HWA features are as follows:

- Supports 2^N ($N=2,3,... 10$) complex FFT of any length input and up to 1024 times adaptive gain scaling, including Rectangular Windows, Hanning Windows, Hamming Windows, Blackman Windows and etc;
- Supports hardware CFAR up to 512 point clouds;
- Supports built in data pre-processing functions, including DC bias elimination, phase alignment between Rx antennas, decoupling between Rx antennas, phase alignment between Tx antennas, static clutter remove and other functions;
- Supports the chirp accumulation(averaging) mechanism, averaging up to 128 chirps;
- Supports Chirp interference detection, abnormal peak detection on time-domain waveform, and zero setting;
- Data output is configurable with flexible allocation of distance FFT and Doppler FFT points. It supports three types of output waveform: time-domain waveform, distance spectrum, and distance Doppler spectrum.

7.3 2.4GHz Wireless Connectivity Subsystem

The RS6130 includes one 2.4GHz wireless connectivity subsystem, which is designed ready for the major feature advancements of Bluetooth®5 and takes advantage of Bluetooth 5's increased performance capabilities which include long range and high throughput modes. The 802.15.4 PHY and MAC layers are supported native on the RS6130. This allows RS6130 to be used in a wide range of home and industrial sensor network applications as it supports two of the most popular wireless sensor standards in use today, Bluetooth low energy and 802.15.4 derivatives. This adds to the already existing radio support for Bluetooth low energy, Thread/Matter and 2.4GHz for proprietary.

The wireless connectivity subsystem contains one 2.4G RF radio transceiver, BLE5 PHY, 802.15.4 PHY, BLE5 Link Layer and 802.15.4 MAC. And the other relative protocol is driven by CPU system. The system diagram is shown in Figure 7-3 2.4GHz Wireless Connectivity Subsystem Diagram.

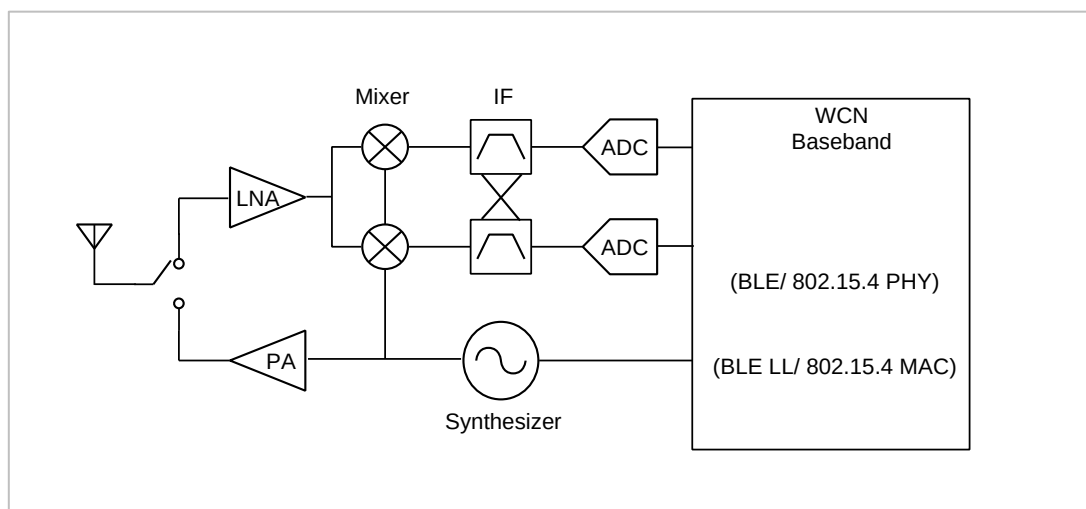


Figure 7-3 2.4GHz Wireless Connectivity Subsystem Diagram

The RF transceiver works in the whole 2.4GHz ISM band, which contains a fully integrate RF synthesizer, a power amplifier, a modulator and a receiver. The RF synthesizer is a Sigma-Delta Fractional-N PLL with LC VCO, while transmit modulation is directly applied to the loop and IQ signal is generate to RX mixer in the receive mode. The receiver includes one Low Noise Amplifier(LNA), IQ mixer and poly-phase IF filter with VGA function, and the Analog-to-Digital Converter(ADC). A switch mode power amplifier is used to optimize PA efficiency, and transmit output power from -20dBm to 10dBm. The RF transceiver also includes several auto calibration machines such as AFC, RX filter tuning, RX offset calibration and PLL bandwidth calibration.

The baseband contains BLE/802.15.4 PHY, BLE Link Layer and 802.15.4 MAC. It supports 125Kbps, 500Kbps, 1Mbps or 2Mbps for BLE5 and 250Kbps for IEEE802.15.4. The baseband also contains hardware logic to perform fast AGC control, access code correction, CRC checking, data whitening, encryption/decryption and frequency hopping logic.

7.4 Application Subsystem

The Application Subsystem is the brain of the chip and controls all the device peripherals and house-keeping activities of the device. It contains 32-bit RISC-V CPU processor, Security Engine, ROM and RAM memory, AHB/APB bus, and associated peripherals and house-keeping components such as DMA and Peripherals (UART, SPI, I²C, GPIO, PWM, and others). The block diagram is shown in Figure 7-5.

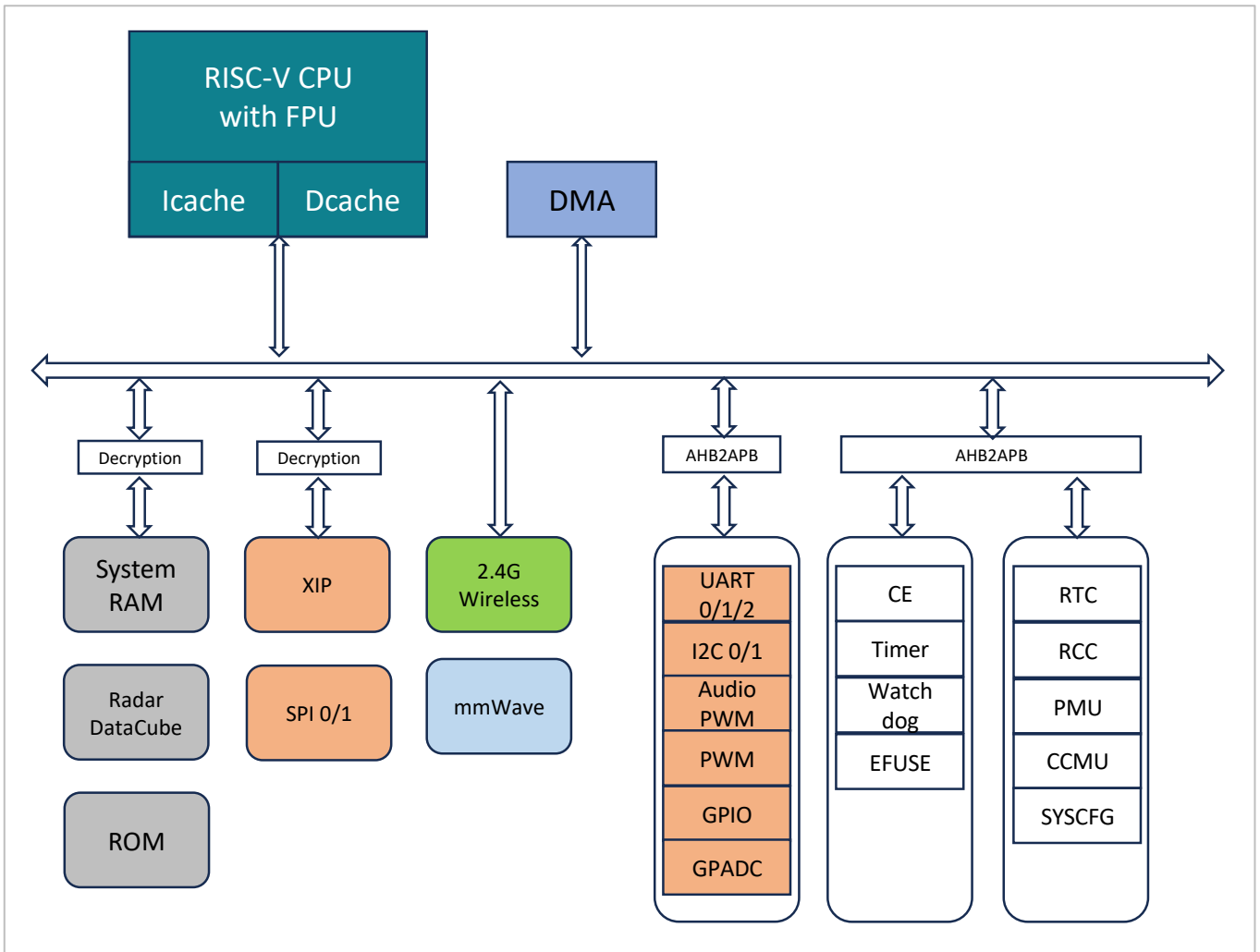


Figure 7-4 Application Subsystem Diagram

7.4.1 CPU Processor

The 32-bit RISC-V CPU processor is clocked with a maximum operating frequency of 128 MHz. User applications executing on this processor control the overall operation of the device, including radar control through well-defined API messages, radar signal processing (assisted by the radar hardware accelerator), 2.4GHz wireless protocol, and peripherals for external interfaces. The processor provides a low-cost platform that meets the needs of minimal memory implementation, reduced pin count, and low power consumption. It has the following features:

- 32-bit RSIC Core, RISC-V 32b/16b variable length ISA
- RV32IMAC compatible
- RV32[F]standard extensions for single-precision floating-point, Compatible with IEEE-754
- Machine mode + user mode
- Improved 5 stage, out-of-order completion
- I & D caches

- Two-level dynamic branch prediction
- 8 channels Physical Memory Protection (PMP) for memory protection features
- Core Local Interrupt Controller (CLIC) to implement interrupt processing
- Support 2k-bit Branch History Table (BHT)
- Support 16-entry Branch Target Buffer (BTB)
- Three 32-bit AMBA3 AHB-Lite interfaces: instruction, data and system bus
- Debug Interfaces support 2-wires SWD and 5-wires JTAG

7.4.2 Security Engine

The RS6130 provides fully security protection for IOT application from boot, key, code and network. It has the following features:

- Secure Boot: ROM supports secure boot to check the integrity and authenticity of the firmware before executing it
- Efuse Key Protection: the secure key in the Efuse is prevent to illegally read
- SRAM and XIP Decryption on the fly: Bus supports Decryption on the fly from external RAM and Flash
- Crypto Engine:
 - AES supports 128, 192, 256 three key lengths and ECB, CBC, CFB, OFB, CTR, GCM five modes
 - SHA1, SHA224 and SHA256
 - RSA supports up to 4096
 - ECC supports up to 521
- Hardware TRNG

7.5 Power and Clock Subsystem

7.5.1 Power Management

A single 1.7~5.5V power supply is required for the RS6130. It could be from an AC-DC converter, USB to supply to 5V or a DC-DC converter to convert higher voltage supply to 3.3V or even lower. It could be from a battery directly too, no matter it is lithium, single 3V button or 2 serial NI-MH battery.

The Power Management Unit (PMU) contains a power clamp, several Low Drop-out Regulators (LDOs), a reference band-gap circuit and temperature sensor. The circuits are optimized for low quiescent current, low drop-out voltage, load regulation, high ripple rejection, and low output noise. The PMU integrates several LDOs for different circuits: RTC LDO, SoC LDO, EXT LDO and INT LDO. They have different operating conditions and features:

- RTC LDO is the main supply only for RTC domain to optimize power consumption at Standby state.

- SoC LDO is the main supply for whole chip digital circuit with programmable voltage from 0.65V to 1.35V to let DVFS operate effectively.
- EXT LDO is main power supply for external device in application, and also can be provide to VDDIO. It has maximum 200mA load current. The output voltage is limited to 3.3V, when VBAT is lower than the value, it will automatically switch to bypass mode to let output voltage follow VBAT.
- INT LDO provide programmable voltage from 0.9V to 2.5V with maximum 600mA load current, for Analog and RF, also SoC LDO input. Normally, make sure VCC voltage is higher than this programmable output voltage setting.

The chip supports to use external DC-DC to further reduce power consumption. After external DC-DC is used, the output voltage on V13 should be at least 0.1V step higher than INT LDO setting voltage to have DC-DC be operating normally.

There are four power domains in the system: RTC domain, AO domain, SoC Switching domain and Radar domain. They mainly used for different scenario to maintain ultra-low power application. We define RS6130 into Active, Idle, Standby and Shutdown power management states to activate different circuit blocks, as shown in Table 7-1.

Table 7-1 Power Management with Block States

Power Mode		CPU	Radar	WCN	EXT /INT LDO	RTC LDO	SoC LDO	DCXO/M SI/DPLL /SPLL	LSI/LSE
Active	All	ON	ON	ON	ON	ON	ON	ON	ON
	CPU	ON	OFF	OFF	ON	ON	ON	OFF/ON/ OFF/ON	ON
	Radar	OFF	ON	OFF	ON	ON	ON	ON	ON
	WCN	ON	OFF	ON	ON	ON	ON	ON/ON/ OFF/OFF	ON
Idle	All	OFF	OFF	OFF	ON	ON	ON	ON	ON
	CPU	OFF	OFF	OFF	ON	ON	ON	OFF/ON/ OFF/ON	ON
	Radar	OFF	OFF	OFF	ON	ON	ON	ON/ON/ ON/OFF	ON
	WCN	OFF	OFF	OFF	ON	ON	ON	ON/ON/ OFF/OFF	ON
Standby		OFF	OFF	OFF	ON	ON	OFF	OFF	ON
Shutdown ¹		OFF	OFF	OFF	ON	ON	OFF	OFF	OFF

1. Chip NRST pin is set to low.

7.5.2 Clock

The clock management system can source the system clocks from a range of internal or external high and low frequency oscillators and distribute them to modules based up the subsystem's individual requirements. The chip has two different kinds of clock source: low frequency clock source(LFCLK) and high frequency clock source(HFCLK).

The chip supports two LFCLK clock sources, the 32.768 KHz crystal oscillator(LSE), LSI(250KHz~2MHz, configurable) The 32.768 KHz crystal oscillator requires an external AT-cut quartz crystal to be connected to the LXL1 and LXL2 pins. The LFCLK is switched off by default when the system is powered up. The LFCLK can be started by selecting the preferred clock source in PRCM register. It is used for each subsystem to achieve lower current consumption for different running mode. In addition, the LFCLK is also used in RTC circuit to achieve accuracy timing.

There are two HFCLK, the 32/40MHz crystal oscillator (XTAL) and 32MHz MSI. The 32MHz MSI is enabled automatically when the system is powered up and can be switched off when all subsystems won't use it anymore in some low power modes. The 32/40MHz XTAL is powered on by CPU after application system is on by software, and controlled by digital signal, so we also name it DCXO in application.

When 2.4GHz wireless subsystem is used, we can only use 32MHz XTAL, while only radar subsystem can use 32MHz or 40MHz XTAL. If higher phase noise performance is required, it is recommended to use 40MHz XTAL for radar.

The HFCLK is used to generate the clock source for SPLL, DPLL WCN SYN, which is used to generate the clock sources for radar, wireless, application system and peripherals. The following figure shows the clock system:

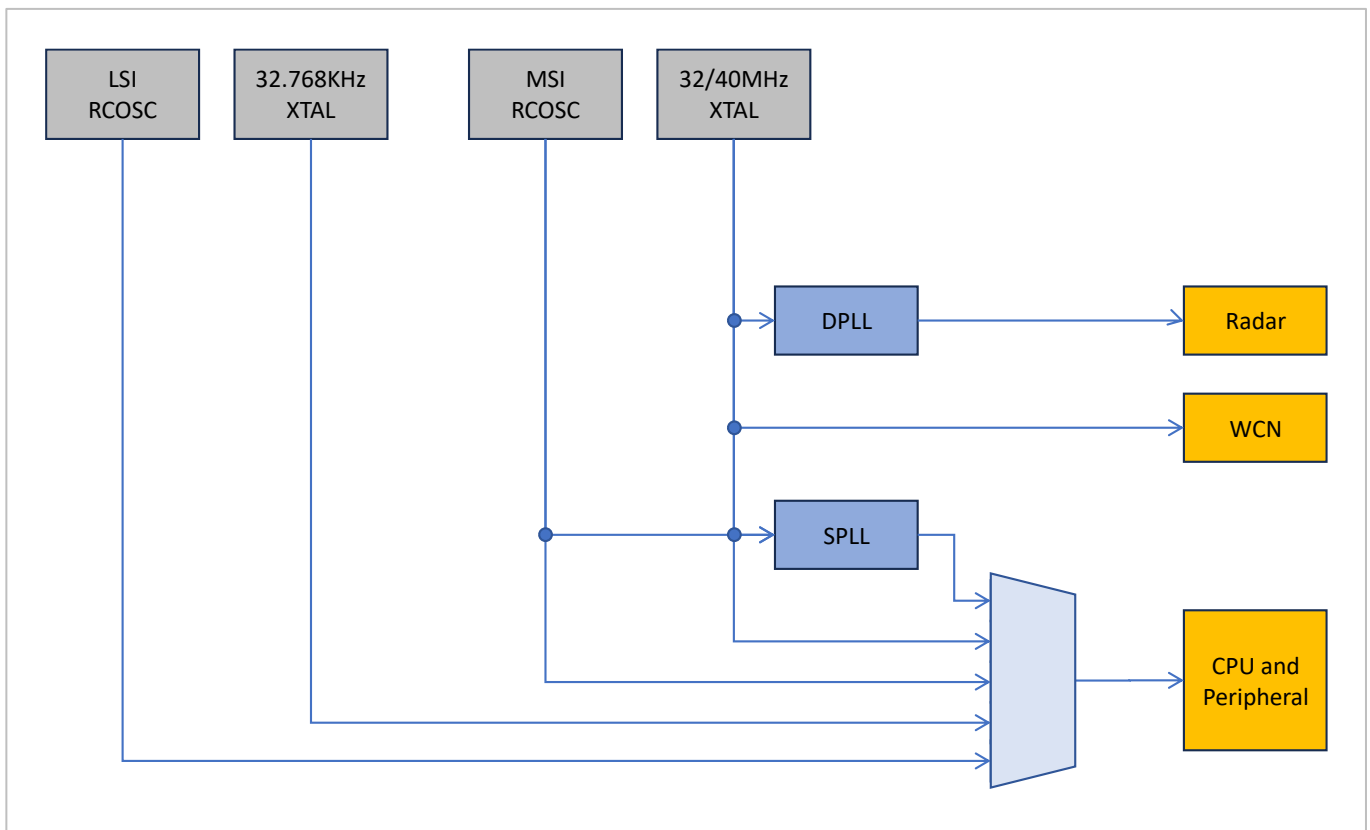


Figure 7-5 Clock System

7.6 Peripherals

7.6.1 GPIO

The RS6130 provides 22 GPIO (General Purpose IO) pins. All ports are mapping out of the device using alternate function multiplexing. The GPIO function can be multiplexed on a multi-function I/O pin by selecting the alternate function in the GPIO Controller Register.

Each GPIO can be configured with the following options:

- Input / Output / Floating(Hi-Z) mode
- Input mode: Pull-up or Pull-down
- Output mode: Active driving@20mA by default, adjustable 20mA per step, up to max 80mA
- Output type: Push-pull or Open-drain
- External Interrupt IO with 5 trigger modes: rising edge, falling edge, double edge, high-level, low-level
- 16 Wakeup IOs can be set to wake system by external interrupt at Standby state
- All IOs can be set to wake system by external interrupt in Idle state

8 IOs are multiplexed with 8 channels ADC. A dedicated internal control signal is used to select between the digital and analog functions.

7.6.2 UART

The RS6130 have three embedded UART controllers: UART2 is used for debug, UART0 and UART1 with auto-flow control are used for communication with external devices. UART0 is used for system boot configuration and firmware upgrade by default. The UART has industry-standard 16550 mode of operation.

Features:

- Compatible with industry-standard 16550
- Support DMA controller interface
- 64-Bytes Transmit and Receive data FIFO
- Support Programmable fractional baud rate
- Support Hardware auto-flow control
- Support baud rate detection
- Support RS-232
- Support clock detection to wake up system from Standby state

7.6.3 QSPI

The RS6130 embeds a Quad Serial Peripheral Interface (QSPI). The QSPI protocol supports half-duplex, full-duplex, synchronous and serial communication with peripheral device. The interface can be configured as master and slave. There are four possible combinations for the serial clock phase and polarity.

Features:

- Clock frequency up to 96MHz
- Support Motorola SPI mode

- Programmable Master or Slave
- Support Standard SPI, Dual SPI and Quad SPI mode
- Programmable Data width size 4 to 32 bits
- 32 words FIFO depth for both Transmit and Receive data
- Programmable sample delay time in master mode
- Programmable Serial Clock Polarity (SCPOL) and Serial Clock Phase (SCPH)
- Support Interrupt and DMA
- Programmable Instruction, Address Length and Wait Cycle in Master mode
- Support clock detection to wake up system from Standby state

7.6.4 I²C

The RS6130 embeds two two-wire serial interfaces (I²C). Each I²C can be configured as master or slave mode. The I²C can operate in standard mode (with data rate 0 to 100 Kb/s), fast mode (with data rates less than or equal to 400 Kb/s), fast mode plus (with data rates less than or equal to 1000 Kb/s).

Features:

- Programmable Master or Slave
- Support three speeds: Standard mode (0 to 100 Kb/s), Fast mode (≤ 400 Kb/s), Fast mode plus (≤ 1000 Kb/s)
- Support 7- or 10bit addressing
- Support Multi-master system
- 8 bytes FIFO depth for Transmit and Receive data
- Programmable SDA hold time
- Support DMA
- Support Address detection to wake up system from Standby mode

7.6.5 PWM

The RS6130 has 4 PWM generators, every PWM generator have a separate counter to output 2 channels. Each PWM generator is a complementary pair output. Only PWM 0/2/4/6 channels are provided with capture mode. The capture detects the rising edge and the falling edge of the signal and calculate capture counter to compare match value.

Features:

- 4 PWM generators, 8 output channels and 4 input capture channels
- Support input capture and interrupt

- Programmable dead-band
- Support pulse, period and complementary pair outputs
- PWM generation with full modulation capability (0-100%)

7.6.6 Audio PWM

The PWM Audio is a high-efficiency mono Class-D Audio Amplifier. The digital audio interface accepts 32 kHz, 44.1 kHz and 48 kHz sample rate with 16-/32-bit data supported for all data formats. The PWM Audio pins is multiplexed with PWM.

Features:

- Support mono mode
- Configurable sample rates: 32 kHz, 44.1 kHz, 48 kHz
- 64 FIFO depth for mono mode
- Support Volume control
- Support interrupt

7.6.7 GPADC

The GPADC contains a 10-channel analog switch, a single end input asynchronous 12-bit SAR ADC. The channels 0 to 7 are used to detect the voltage of the external input, The channel 8 is used to detecting the voltage of VINT, and the channel 9 is dedicated to detecting the voltage of the VCC.

Features:

- 12-bit resolution and 10-bit effective SAR type A/D converter
- 10-channel multiplexer, 8 normal channels and 2 VCC voltage detection channel
- Up to 32 FIFO depth for data
- Support DMA
- Support calibration and compensation
- Internal Vref 1.2V, Support five class of input voltage attenuation: 2/3, 1/2, 1/3, 1/5, 1/9. The max voltage is 3.6V
- Maximum sampling frequency: 2 MHz
- Programmable oversampling: 16, 32, 64, 128
- Support one channel 4096 oversampling
- Support compare and data interrupt
- Support four operation modes: single mode, single-cycle mode, continuous mode, burst mode

- Support hardware triggers (internal timer periodic event or WIO input event) to start conversion in STANBY mode. It generates an interrupt to wake system when data is over the range of the comparison window.

8 Package Description

8.1 Package Outline

The RS6130 chip-set is housed in the FCCSP package with bump balls of 350µm diameter and standoff of 280 µm. The package size is (8.8 x 5.8 x 0.68) mm³ with a general ball pitch of 650 µm.

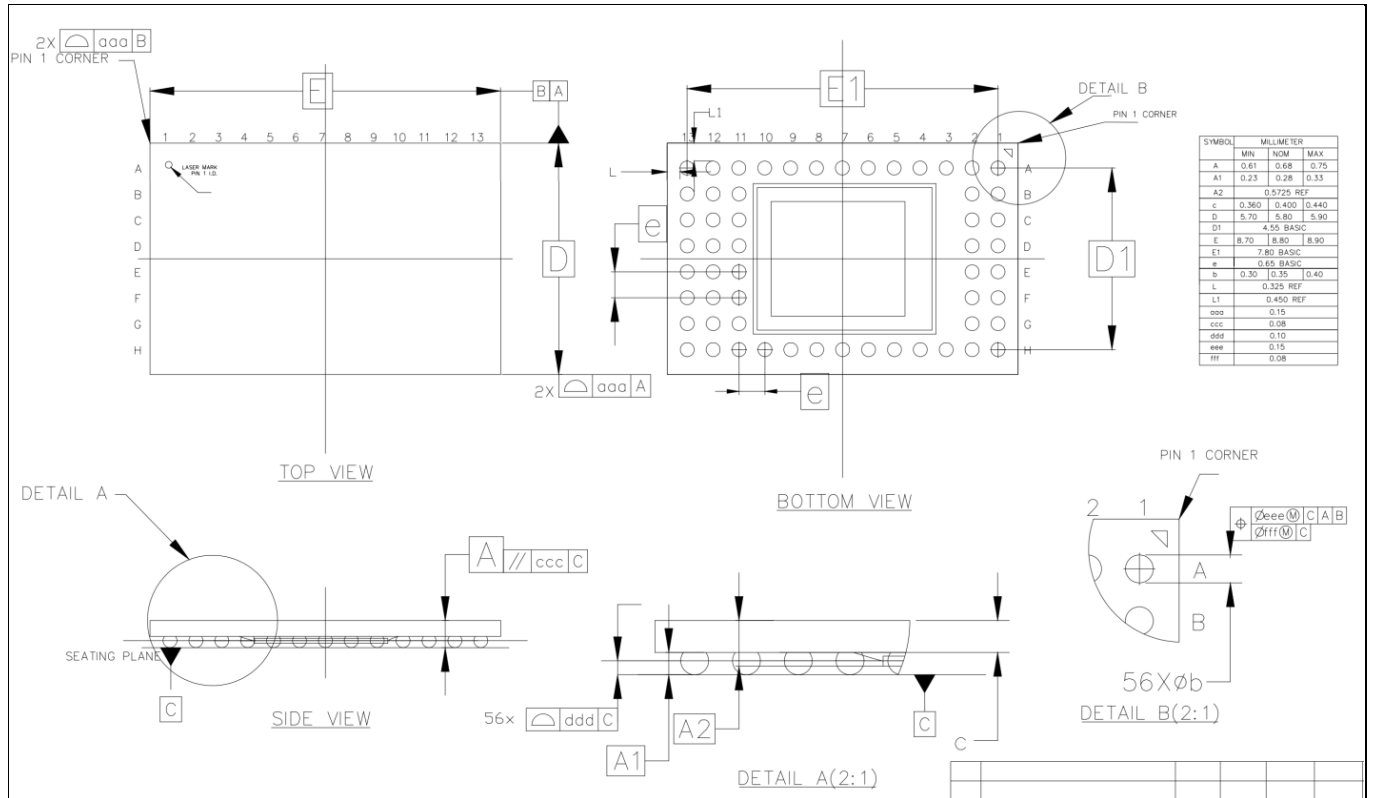


Figure 8-1 Package Outline Diagram

8.2 PCB Land Pattern

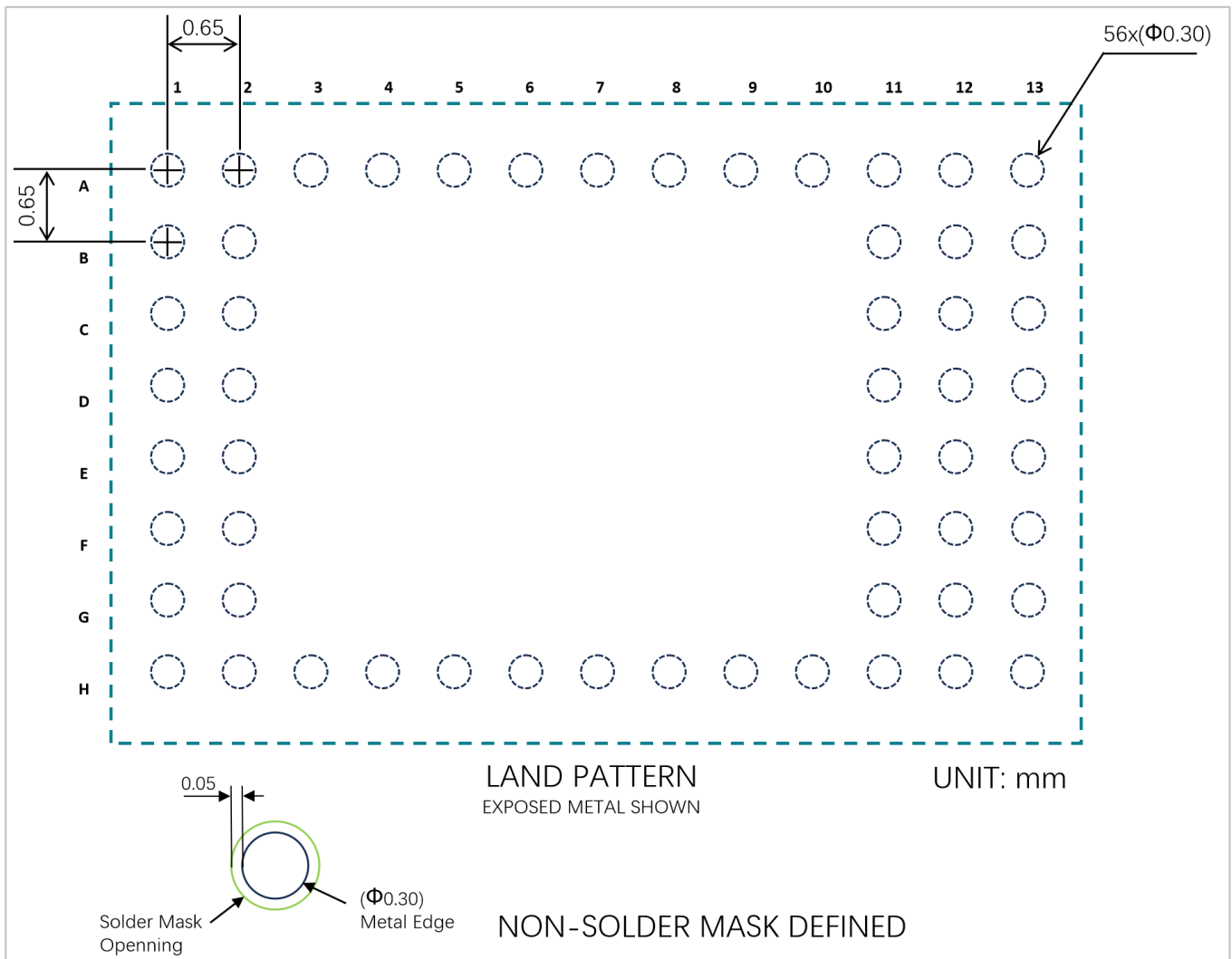


Figure 8-2 PCB Land Pattern Diagram

8.3 Package Thermal Property

Table 8-1 Thermal Property

Thermal Parameters		°C/W
$R\theta_{JC}$	Junction-to-case	TBD
$R\theta_{JB}$	Junction-to-board	TBD
$R\theta_{JA}$	Junction-to-free air	TBD
Ψ_{iJA}	Junction-to-package top	TBD
Ψ_{iJB}	Junction-to-board	TBD

9 Application Circuit

9.1 Schematic Notes

- **POWER**

- **OPTION A:** “DCDC” section will help to reduce power consumption compared to using RS6130 Inner-LDO, if application don’t care this, can delete it to save BOM cost.
- VCC and VDDIO are powered by different external power source, however, when VCC is below 3.6V, VCC and VDDIO could share the same power source.

- **BLE**

- **OPTION B:** “32.768KHz” section is used for BLE system, if application doesn’t contain BLE, can delete it.
- **OPTION C:** “BLE ANTENNA” section is used for BLE system, if application doesn’t contain BLE, can delete it.

- **EXTERNAL CONNECT**

HOST MCU could control the radar through any interface in UART0/I2C0/SPI0, and UART0 is used by default.

PA6 and PA11 is used as Trigger output signal, usually select one.

PB0 is used as firmware download control signal, if user want to update firmware, should connect it.

It is strongly recommended to reserve PA0, PB0, and PB1 as test points, as these three signals will be used for noise testing, it will be convenient for radar performance analysis.

- **OPTION D:** CONNECT By I2C0
- **OPTION E:** CONNECT By SPI0

9.2 Reference Schematic

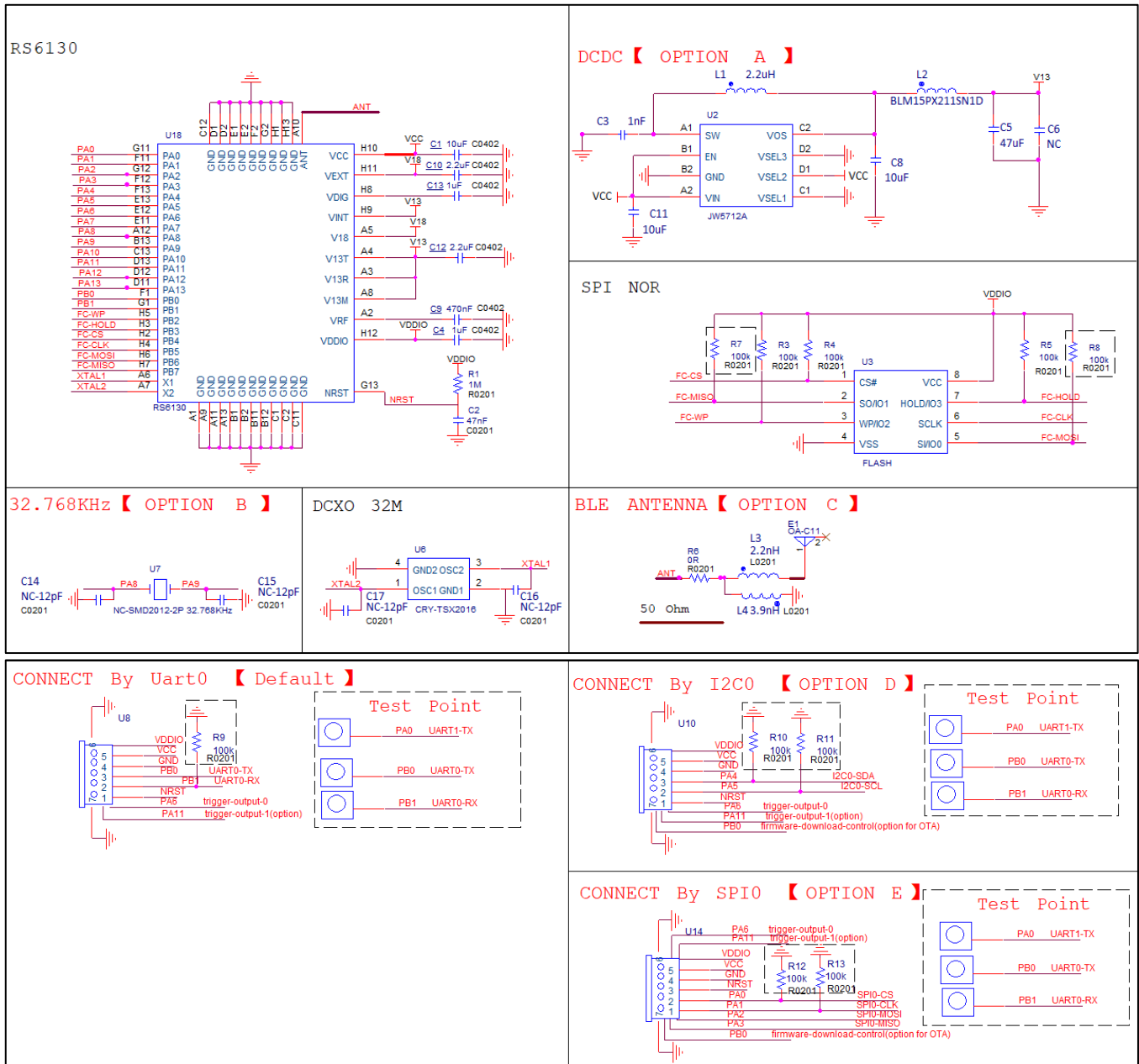


Figure 9-1 Reference Schematic

10 Silkscreen

The RS6130 package contains an antenna design, so silkscreen information is printed on the bottom of the chip.

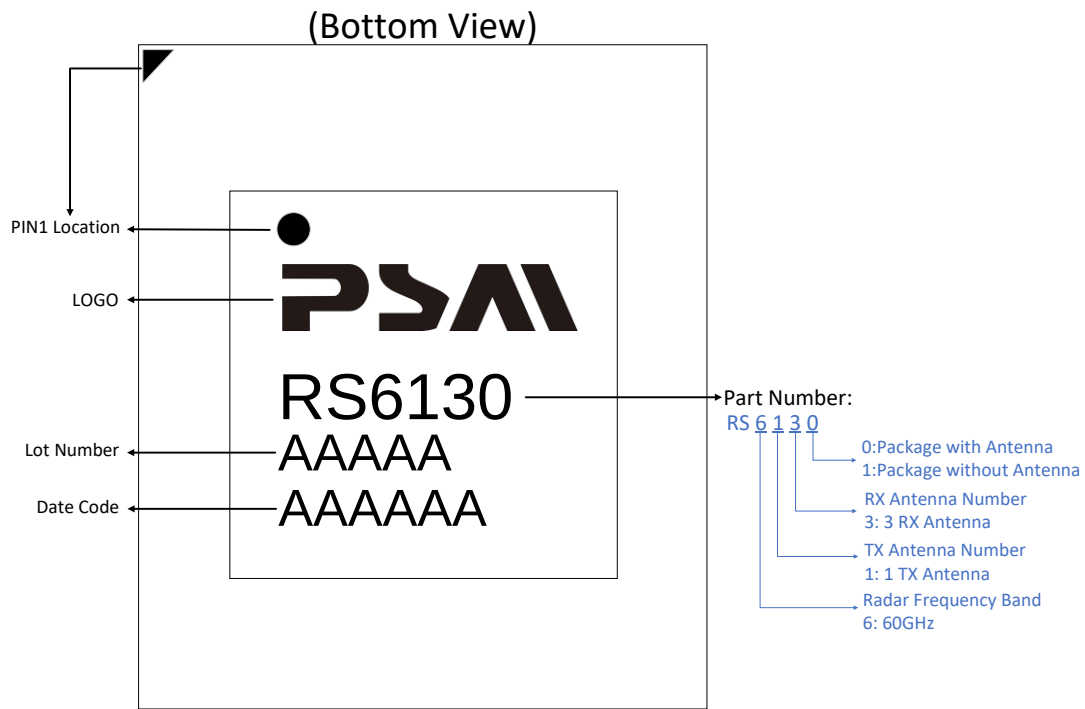


Figure 10-1 Silkscreen

11 Shipping package

Unit: mm

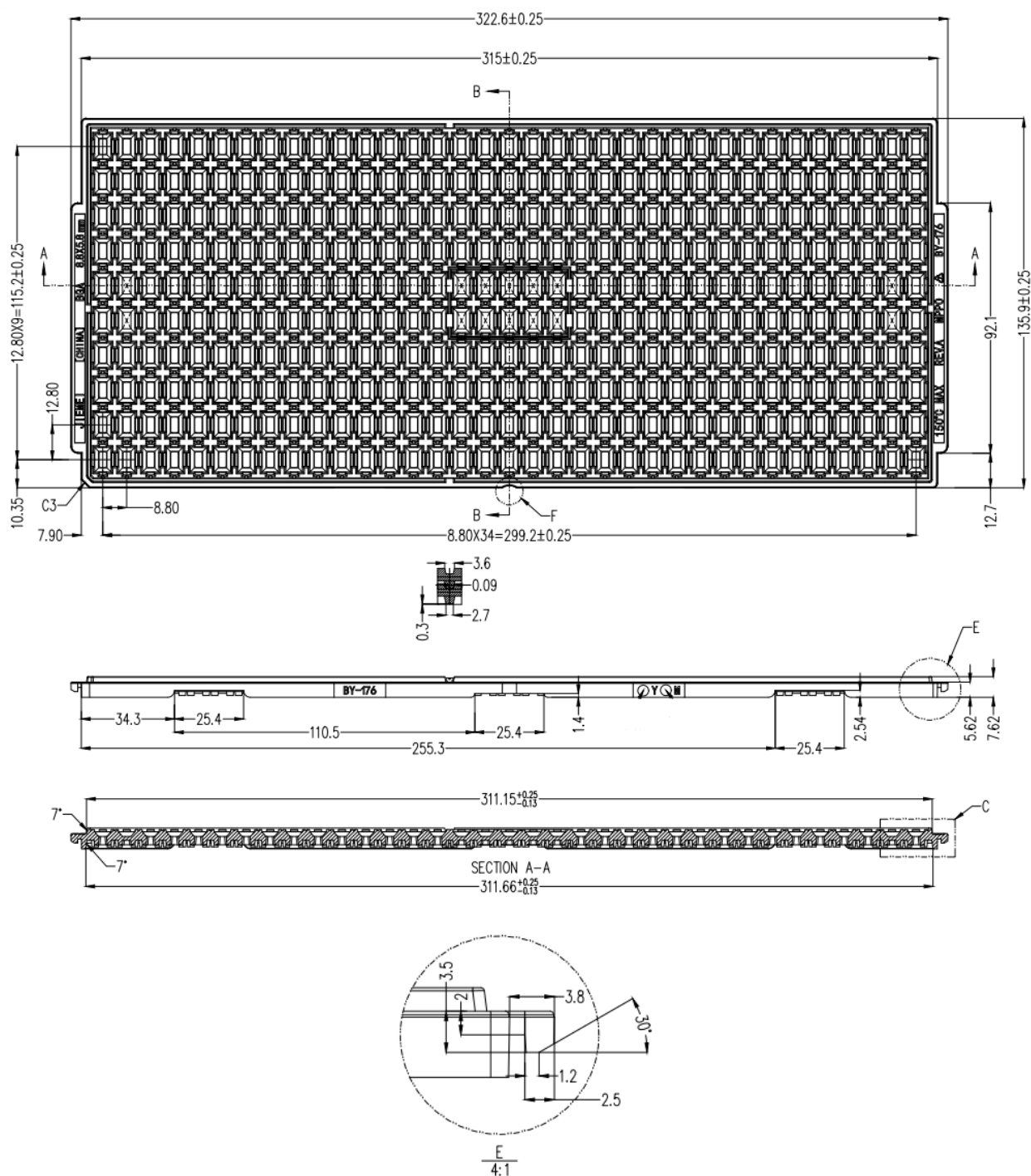


Figure 11-1 Shipping package 1

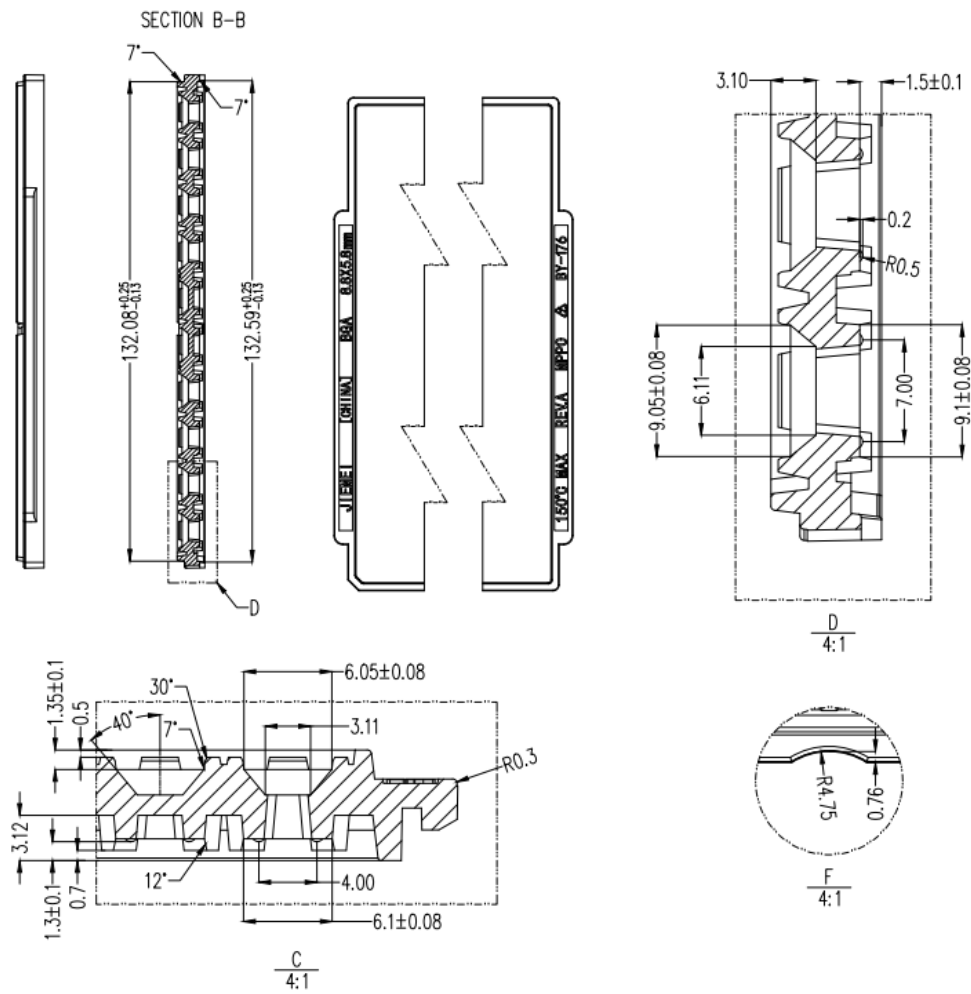


Figure 11-2 Shipping package 2

12 Reflow Soldering

- MSL-3: The moisture sensitivity level is 3.
- Suggested peak reflow temperature is 240°C~260°C, not exceed 260°C, duration can not exceed 15 seconds. The recommended reflow soldering heat curve (lead-free reflow soldering) and related parameters are as follows:

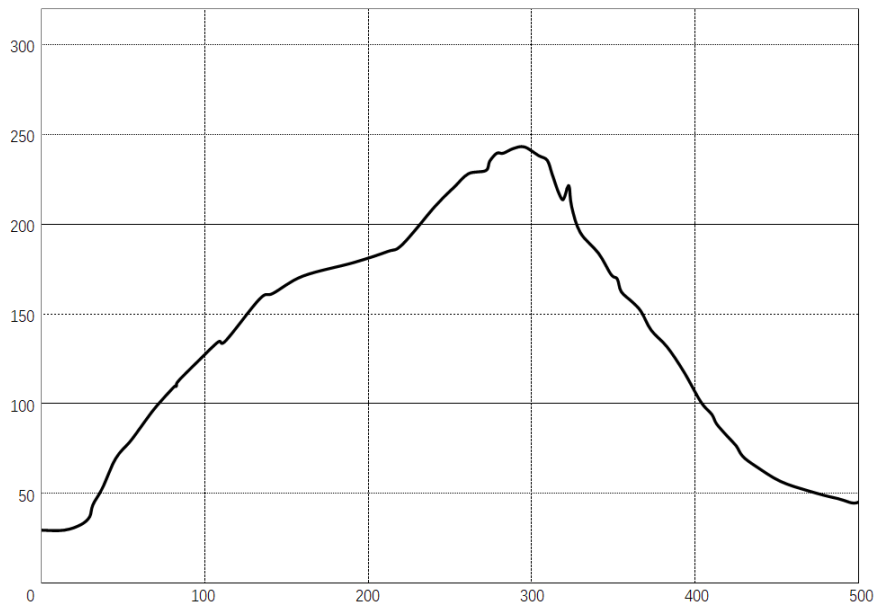


Figure 12-1 Recommended reflow soldering heat curve

Table 12-1 Recommended thermal distribution parameters

Parameter	Recommended value
Max. slope for heating up	2.0~4.0°C/sec
Max. slope for cooling down	1.0~5.0°C/sec
Soak time (150°C~ 200°C)	60~120 sec
Reflow time (over 220°C)	30~90 sec
Max. temperature	240°C~260°C

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